

Database Schema Specification

Student OS Documentation — File 08

Core Entities

User: id, name, email, password/auth reference, role, createdAt, updatedAt.

StudentProfile: userId, program, semester, section, batch, academicYear.

Subject/Course: id, code, name, credits, faculty reference where applicable.

TimetableEntry: id, subjectId, dayOfWeek, startTime, endTime, room, faculty, type.

Task: id, userId, subjectId, title, description, dueAt, priority, category, status, completedAt.

Event: id, userId, type, title, description, startAt, endAt, relatedEntity.

AttendanceRecord: id, userId, subjectId, date/class reference, attended/status.

Exam: id, userId, subjectId, title, examAt, room, syllabus, preparationProgress, status.

Note: id, userId, subjectId, title, content, createdAt, updatedAt.

Notification: id, userId, type, title, message, readAt, scheduledAt, relatedEntity.

NotificationPreference: userId, type, enabled, timing.

StudySession: id, userId, subjectId, startedAt, endedAt, duration.

Attachment: id, ownerId, entityType, entityId, storageKey, filename, size, mimeType.

Relationships

User has one StudentProfile.

User has many Tasks, Events, Exams, Notes, Notifications, AttendanceRecords and StudySessions.

Subject has many TimetableEntries, AttendanceRecords, Exams, Tasks and Notes.

Related entities should use stable IDs and foreign-key constraints where supported.

Data Rules

Use UTC or a clearly defined timezone strategy for timestamps.

Index userId and frequently filtered date/status fields.

Use soft deletion only where business/history requirements justify it.

Never store plaintext passwords.

Validate ownership before returning user-specific records.

Migration Rules

Schema changes must be versioned and reversible where practical. Seed data must be clearly marked as development/demo data.